Algebraic Topology

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Michaelmas 2023*

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Lecture 1

1 Introduction

Let X, Y be topological spaces, $f, g : X \to Y$ be continuous functions. Note we will use the terminology 'map := continuous function'. We say that f *is homotopic to* g, written $f \simeq g$, if there exists a map $F : X \times I \to Y$, such that

$$F|_{X\times\{0\}} = f \quad \textit{and} \quad F|_{X\times\{1\}} = g$$

Here $I = [0, 1]$ carries its Euclidean topology. An exercise (sheet 1) is that this defines an equivalence relation on the set of maps $X \to Y$.

Definition 1.0.1 (homotopy equivalence)
We say that $f : X \to Y$ is a *homotopy equivalence* if there exists a function $g : Y \to X$ such that $f \circ g \simeq \mathrm{id}_Y$ and $g \circ f \simeq \mathrm{id}_X$. We say that g is a *homotopy inverse* to f.

The same exercise on sheet 1 shows that homotopy equivalence is an equivalence relation on topological spaces.

Example 1.0.2
1. If $f : X \to Y$ is a homeomorphism, then f is also a homotopy equivalence.
2. $i : \{0\} \hookrightarrow \mathbb{R}^n$ is a homotopy equivance.
3. The inclusion $i : S^{n-1} \hookrightarrow \mathbb{R}^n \setminus 0$ is a homotopy equivalence.

Definition 1.0.3
A space X is *contractible* if it is homotopy equivalent to a point.

*Based on lectures by Ivan Smith. Last updated October 21, 2023.

Idea of Algebraic Topology: Study spaces up to homotopy equivalence. Broadly, we are interested in "connectivity properties" of topological spaces.

Example 1.0.4

1. We say a space X is *path-connected* if any two maps $\{*\} \to X$ are homotopic. For example, $\mathbb{R}$ is path connected, $\mathbb{R} \setminus \{0\}$ is not. A corollary of this is the intermediate value theorem.
2. We say a path connected space X is *simply connected* if every map $f : S^1 \to X$ is homotopic to a constant map. Equivalently, every two maps $S^1 \to X$ are homotopic. Or equivalently (again) every continuous map $S^1 \to X$ extends to a continuous map $D^2 \to X$. For example, $\mathbb{R}^2$ is simply connected, by $\mathbb{R}^2 \setminus \{0\}$ is not. If $\gamma : S^1 \to \mathbb{R}^2 \setminus \{0\}$ is continuous, it has a *winding number*, or *degree* $\deg(\gamma) \in \mathbb{Z}$, such that
   - It is invariant under homotopy,
   - If $\gamma_n(t) = \exp(2n\pi it)$, then $\deg(\gamma_n) = n$

   In particular, taking $n = 0$ gives the constant map. Therefore, a constant map has degree 0. A corollary of this is the fundamental theorem of algebra.
3. We say a path connected space X is *k-connected*, if for all $i \leq k$, every map $S^i \to X$ is homotopic to a constant map. In this case, $\mathbb{R}^n$ is $(n-1)$-connected, but $\mathbb{R}^n \setminus 0$ is not. More precisely, maps from $S^{n-1} \to \mathbb{R}^n \setminus 0$ have a degree $\deg(f) \in Z$, such that degree is a homotopy invariant, and the degree of the inclusion map is 1, and the degree of the constant map is 0. A corollary of this is the Brouwer fixed point theorem.

Lecture 2

2 (Co)homology

2.1 Co(chain) complexes

We will define invariants of topological spaces in two steps:

(a) We associate to X a (co)chain complex,

(b) We take the (co)homology of the complex

The topology enters in step (a), whereas (b) is just homological algebra.

Definition 2.1.1 (chain complex)

A *chain complex*@ (C_*, ∂) consists of a sequence $\{C_i\}_{i\in\mathbb{Z}}$ of abelian groups, along with *boundary homomorphisms* $\partial_n : C_n \to C_{n-1}$ for all n, satisfying $\partial_n \circ \partial_{n+1} = 0$. We also write this conditions as $\partial^2 = 0$.

Definition 2.1.2 (homology group)

The *i-th homology group* of a chain complex (C_*, ∂) is

$$\mathrm{H}_i(C_*, \partial) = \frac{\ker(\partial_i : C_i \to C_{i-1})}{\mathrm{im}(\partial_{i+1} : C_{i+1} \to C_i)}$$

We write

$$\mathrm{H}_*(C_*, \partial) = \bigoplus_{i\in\mathbb{Z}} \mathrm{H}_i(C_*, \partial)$$

Definition 2.1.3 (cochain complex)

A *cochain complex* (C^*, ∂) consists of a sequence $\{C^i\}_{i\in\mathbb{Z}}$ of abelian groups, along with *boundary ho-*

momorphisms $\partial^n : C^n \to C^{n+1}$ for all n, satisfying $\partial^{n+1} \circ \partial^n = 0$. We also write this conditions as $\partial^2 = 0$.

Definition 2.1.4 (cohomology group)

The *i-th cohomology group* of a cochain complex (C^*, ∂) is

$$\mathrm{H}^i(C^*, \partial) = \frac{\ker(\partial^i : C^i \to C^{i+1})}{\mathrm{im}(\partial^{i-1} : C^{i-1} \to C^i)}$$

We write

$$\mathrm{H}^*(C^*, \partial) = \bigoplus_{i \in \mathbb{Z}} \mathrm{H}_i(C^*, \partial)$$

Notation 2.1.5. Elements of $\ker(\partial)$ are called *cycles* (in a chain complex), *cocycles* (in a cochain complex). Elements of $\mathrm{im}(\partial)$ are called *boundaries* (and *coboundary* resp.).

Elements of H_* are called *homology classes*, and elements of H^* are called *cohomology classes*.

We will call ∂ the *differential*.

Definition 2.1.6 (chain map)

Given chain complex (C_*, ∂), (D_*, ∂), a *chain map* $f : C_* \to D_*$ comprises group homomorphism $f_i : C_i \to D_i$, such that

$$\begin{array}{ccc} C_i & \xrightarrow{f_i} & D_i \\ \downarrow{\scriptstyle \partial} & & \downarrow{\scriptstyle \partial} \\ C_{i-1} & \xrightarrow[f_{i-1}]{} & D_{i-1} \end{array}$$

commutes.

Lemma 2.1.7. A chain map $f : C_* \to D_*$ induces homomorphisms $f_* : \mathrm{H}_i(C_*) \to \mathrm{H}_i(D_*)$.

Proof. Let $a \in \mathrm{H}_i(C_*)$, and we can choose a cycle $\alpha \in C_i(C_*)$, with $\partial\alpha = 0$. In this case,

$$\partial f_i(\alpha) = f_{i-1}(\partial\alpha) = f_{i-1}(0) = 0$$

So $f_i(\alpha) \in D_i$ is a cycle. We set $f_*(a) = [f_i(\alpha)] \in \mathrm{H}_i(D_*)$. Next, we need to show that this is independent of choices. Suppose $a = [\alpha] = [\alpha']$. Then $\alpha - \alpha' \in \mathrm{im}(\partial)$. Therefore, we can write $\alpha - \alpha' = \partial\tau$. In this case,

$$f_i(\alpha) - f_i(\alpha') = f_i(\partial\tau) = \partial f_{i+1}(\tau) \in \mathrm{im}(\partial)$$

So we conclude $[f_i(\alpha)] = [f_i(\alpha')] \in \mathrm{H}_i(D_*)$. It is clear that f_* defines a homomorphism. □

Correspondingly, we can define *cochain maps* $C^* \to D^*$, that is, $f^i : C^i \to D^i$ such that

$$\begin{array}{ccc} C^{i+1} & \xrightarrow{f^{i+1}} & D^{i+1} \\ \uparrow{\scriptstyle \partial} & & \uparrow{\scriptstyle \partial} \\ C^i & \xrightarrow[f^i]{} & D^i \end{array}$$

commutes.

Exercise: This construction is functorial. That is, $(\mathrm{id}_{C_*})_* = \mathrm{id}_{\mathrm{H}_*}$, and if we have $f : C_* \to D_*$, $g : D_* \to E_*$, then $(g \circ f)_* = g_* \circ f_*$.

Goal: To associate to a topological space X (co)chain complex $C_*(X)$ and $C^*(X)$. The definition works for any X, but the theories are better behaved for "nicer" X.

2.2 Singular (co)homology

We will develop singular (co)homology.

Definition 2.2.1 (standard simplex, face)

The *standard simplex* is

$$\Delta^n = \left\{ (t_0, \dots, t_n) \in \mathbb{R}^{n+1} \mid t_i \geq 0, \sum_i t_i = 1 \right\}$$

The i-th face Δ^n is

$$\Delta^n_i = \Delta^n \cap \{t_i = 0\}$$

and we have a canonical homeomorphism

$$\begin{aligned} \delta_i : \Delta^{n-1} &\to \Delta^{n-1}_i \\ (t_0, \dots, t_{n-1}) &\mapsto (t_0, \dots, t_{i-1}, 0, t_i, \dots, t_{n-1}) \end{aligned}$$

Definition 2.2.2 (singular simplex, singular chain complex)

If X is a topological space, a *singular n-simplex* in X is a continuous map $\sigma : \Delta^n \to X$.

The *singular chain complex* is given by the the free abelian groups

$$C_i(X) = \bigoplus_{\sigma \text{ singular } n\text{-simplex}} \mathbb{Z}\sigma$$

We define the boundary map $C_i(X) \to C_{i-1}(X)$ via

$$\partial\sigma = \sum_{j=0}^{i} (-1)^j \sigma|_{\Delta^i_j}$$

where

$$\sigma|_{\Delta^i_j} := \sigma \circ \delta_j : \Delta^{n-1} \to X$$

Note if $\{v_i\}_{i=0}^n$ are $n+1$ ordered points in $\mathbb{R}^{n+1}$, and if $\{v_i - v_0\}_{i=1}^n$ are linearly independent, then the convex hull of $v_0, \dots, v_n$ is an n-simplex, which we will call

$$[v_0, \dots, v_n]$$

which is given by the map

$$\begin{aligned} \Delta^n &\to \mathbb{R}^{n+1} \\ t &\mapsto \sum_i t_i v_i \end{aligned}$$

We orient the edges of the standard simplex (and thus any simplex $[v_0, \dots, v_n]$) by saying that $v_i < v_j$ if $i < j$.

More concretely, for $n = 2$, we have the oriented edges $v_0 \to v_1, v_0 \to v_2, v_1 \to v_2$.

Lemma 2.2.3. $\partial^2 = 0$.

Proof. If σ is an n-simplex, defined on $[v_0, \dots, v_n]$, then

$$\partial\sigma = \sum_j (-1)^j \sigma|_{[v_0, \dots, \widehat{v_j}, \dots, v_n]}$$

and so

$$\partial(\partial\sigma) = \sum_{i<j} (-1)^i (-1)^j \sigma|_{[v_0, \dots, \widehat{v_i}, \dots, \widehat{v_j}, \dots, v_n]} + \sum_{i>j} (-1)^{i+1} -1^j \sigma|_{[v_0, \dots, \widehat{v_j}, \dots, \widehat{v_i}, \dots, v_n]} = 0$$

since the signs cancel. □

Definition 2.2.4

The *singular homology groups of* X, $\mathrm{H}_i(X)$, are the homology groups of the singular chain complex.

Remark 2.2.5. Since all we used is the topology on X, $\mathrm{H}_i(X)$ is clearly a topological (read homeomorphism) invariant of X.

Moreover, if $f : X \to Y$ is continuous, $\sigma : \Delta^n \to X$ is an n-simplex on X, then $f \circ \sigma$ is an n-simplex on Y. The fact that

$$(f \circ \sigma) \circ \delta_j = f \circ (\sigma \circ \delta_j)$$

shows that f induces a chain map.

Lecture 3

Definition 2.2.6 (singular cochain complex)

The *singular cochain complex* of X has

$$C^i(X) = \mathrm{Hom}_{\mathbb{Z}}(C_i(X), \mathbb{Z})$$

and the boundary operator

$$\partial^* \psi(\sigma) = \psi(\partial \sigma)$$

for $\psi \in C^i(X)$, $\sigma \in C_{i+1}(X)$. That is, it is adjoint to $\partial : C_i(X) \to C_{i-1}(X)$.

Since $\partial^2 = 0$, $(\partial^*)^2 = 0$. The *singular cohomology* is

$$\mathrm{H}^*(X) = \bigoplus_{i \geq 0} \mathrm{H}^i(C^*, \partial^*)$$

If $f : X \to Y$ continuous, then f induces a chain map $f_* : C_*(X) \to C_*(Y)$ via $f_*(\sigma) = f \circ \sigma$, satisfying $\partial f_* = f_* \partial$. Similarly, $f : X \to Y$ defines a *pullback map* $f^* : C^*(Y) \to C^*(X)$, given by

$$f^*(\psi)(\tau) = \psi(f_*(\tau))$$

Again, we have that

$$\partial^* f^*(\psi)(\tau) = f^* \psi(\partial \tau) = \psi(f_* \partial \tau) = \psi(\partial f_* \tau) = (\partial^* \psi)(f_* \tau) = f^*(\partial^* \psi)(\tau)$$

So f^* is a cochain map, i.e. $\partial^* f^* = f^* \partial^*$. Therefore, we have an induced homomorphism on cohomology,

$$f^* : \mathrm{H}^*(Y) \to \mathrm{H}^*(X)$$

Warning: by definition,

$$C^i(X) = \mathrm{Hom}_{\mathbb{Z}}(C_i(X), \mathbb{Z})$$

but it is not true (in general) that

$$\mathrm{H}^i(X) = \mathrm{Hom}_{\mathbb{Z}}(\mathrm{H}_i(X), \mathbb{Z})$$

2.2.1 Basic computations

Lemma 2.2.7 (homology of a point). Let $X = \{\bullet\}$. Then

$$\mathrm{H}_n(X) = \begin{cases} \mathbb{Z} & n = 0 \\ 0 & \text{otherwise} \end{cases}$$

Proof. By definition, $C_i(X)$ is the free abelian groups on continuous map $\Delta^i \to \{\bullet\}$. But there is only one such map, namely the constant map (for $i \geq 0$). Hence the chain complex looks like

$$\cdots \longrightarrow C_2(X) \longrightarrow C_1(X) \longrightarrow C_0(X) \longrightarrow 0$$

where each $C_i(X)$ is $\mathbb{Z}$. Computing the boundary maps,

$$\partial(\sigma_n) = \begin{cases} \sigma_{n-1} & \text{if } n \text{ is even} \\ 0 & \text{if } n \text{ is odd} \end{cases}$$

Hence the chain complex is

$$\cdots \longrightarrow \mathbb{Z} \xrightarrow{\text{id}} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{\text{id}} \mathbb{Z} \xrightarrow{0} 0$$

We can then compute the homology groups explicity. □

Lemma 2.2.8. If we write

$$X = \bigsqcup_{\alpha \in A} X_\alpha$$

as a disjoint union of path components, then

$$\mathrm{H}_i(X) = \bigoplus_{\alpha \in A} \mathrm{H}_i(X_\alpha)$$

Moreover, if X is path connected (and nonempty), then $\mathrm{H}_0(X) = \mathbb{Z}$.

Proof. Since Δ^i is path connected, the image of a continuous map $\sigma : \Delta^i \to X$ must lie within some X_α. Moreover, so does the image of all of the boundary faces $\sigma \circ \delta_j$. So in fact

$$(C_i(X), \partial) = \left(\bigoplus_{\alpha \in A} C_i(X_\alpha), \partial \right)$$

Note that a general element of $C_i(X)$ is a finite linear combination of i-simplicies in X. So its image can only meet finitely many of the X_α.

Now suppose X is path connected. Define a function

$$\begin{aligned} \varepsilon : C_0(X) &\to \mathbb{Z} \\ \sum n_i \sigma_i &\mapsto \sum n_i \end{aligned}$$

Since X is nonempty, ε is surjective. On the other hand, if $\tau \in C_1(X)$, then $\varepsilon(\tau(1) - \tau(0)) = 0$. So by linearity,

$$\mathrm{im}(\partial : C_1(X) \to C_0(X)) \subseteq \ker(\varepsilon)$$

Recall

$$\mathrm{H}_0(X) = \frac{C_0(X)}{\mathrm{im}(\partial : C_1(X) \to C_0(X))}$$

and so ε descends to a map $\mathrm{H}_0(X) \to \mathbb{Z}$. Now we will use the fact that X is path connected. Suppose

$$\sum n_i \sigma_i \in \ker(\varepsilon)$$

and fix a base point $p \in X$. Since the σ_i are 0-simplicies, they correspond to points in X. Moreover, since X is path connected, we can choose paths

$$\tau_i : \Delta^1 \to X$$

such that $\tau_i(1) = \sigma_i$ and $\tau_i(0) = p$. In this case,

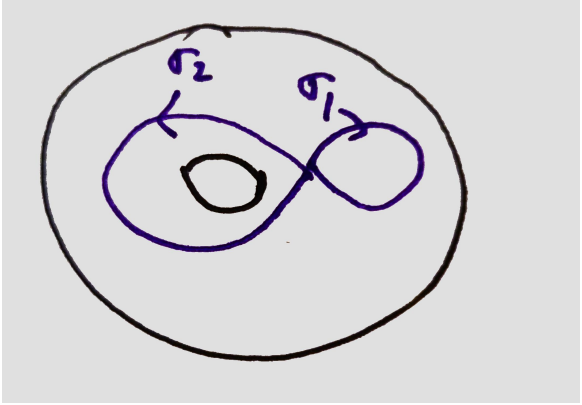
$$\partial \left(\sum n_i \tau_i \right) = \sum n_i \partial \tau_i = \sum n_i \sigma_i - (\sum n_i) p$$

But we assumed $\sum n_i = 0$, and so $\ker(\varepsilon) \subseteq \mathrm{im}(\partial : C_1(X) \to C_0(X))$. This then gives the required result, as the induced map

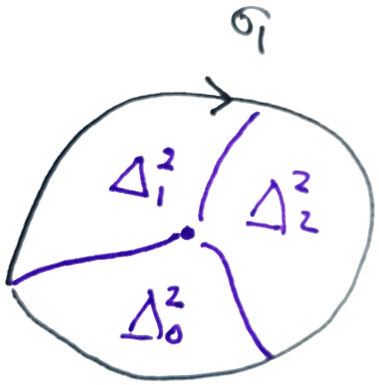
$$H_0(X) \to \mathbb{Z}$$

is then an isomorphism. □

Informal conjecture: We can't compute anything else.
Therefore, we need to develop some more structure about (co)homology.
What are we doing? Consider the case of X being an annulus in $\mathbb{R}^2$. We have 1-simplices



(σ_1 null-homotopic, σ_2 winds around the origin)
In this case, $\partial\sigma_1 = \partial\sigma_2 = 0$, and so they define homology classes.



Moreover, $[\partial\sigma_1] = 0$, since we have triangles τ_0, τ_1, τ_2 such that

$$\partial(\tau_0 + \tau_1 + \tau_2) = \sigma_1$$

From this intuition, $[\sigma_2] \neq 0$. But to show even this, we need some structure theorems.

Lecture 4

2.3 Fundamental properties

(Co)homology is useful by virtue of various structural properties.

Theorem 2.3.1 (homotopy invariance). If $f, g : X \to Y$ are homotopic, then the induced maps f_*, g_* on homology, and the induced maps f^*, g^* on cohomology agree.

Corollary 2.3.2. If $X \simeq Y$, then $H_*(X) \simeq Y$, and if f is a homotopy equivalence, then f_* induces the isomorphism. For cohomology we have the similar statement, but with f^*.

Proof of the corollary. $X \simeq Y$ via f is saying that there exists $g : Y \to X$, such that $g \circ f \simeq \mathrm{id}_X$, and $f \circ g \simeq \mathrm{id}_Y$. That is,

$$g_* \circ f_* = \mathrm{id} \quad \text{and} \quad f_* \circ g_* = \mathrm{id}$$

□

We think of this as saying that (co)homology is "insensitive to inessential deformations".

Example 2.3.3

Recall $\{0\} \hookrightarrow \mathbb{R}^n$ is a homotopy equivalence, and so

$$\mathrm{H}_*(\mathbb{R}^n) = \begin{cases} \mathbb{Z} & * = 0 \\ 0 & \text{otherwise} \end{cases}$$

The other key structural property relies on some homological algebra.

Definition 2.3.4 (exact sequence)

An *exact sequence* is a chain complex with $H_*(C_*, \partial) = 0$. That is,

$$\mathrm{im}(\partial : C_{n+1} \to C_n) = \ker(\partial : C_n \to C_{n-1})$$

Similarly, a cochain complex is *exact* if $H^*(C^*, \partial) = 0$.
If A, B, C are abelian groups, and we have

$$A \xrightarrow{\alpha} B \xrightarrow{\beta} C$$

Then this sequence is *exact at* B if $\ker(\beta) = \mathrm{im}(\alpha)$.

Definition 2.3.5 (short exact sequence)

A *short exact sequence* is an exact sequence of the form

$$0 \longrightarrow A \xrightarrow{\alpha} B \xrightarrow{\beta} C \longrightarrow 0$$

which is exact at all places. That is, α is injective, β is surjective, and $\mathrm{im}(\alpha) = \ker(\beta)$.

Example 2.3.6 (i) If

$$0 \longrightarrow A \xrightarrow{\alpha} B \longrightarrow 0$$

is exact, then α is an isomorphism.

(ii) If

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\alpha} G \xrightarrow{\beta} \mathbb{Z}/n \longrightarrow 0$$

is exact, then we could have $G = \mathbb{Z} \oplus \mathbb{Z}/n$, or we could have $G = \mathbb{Z}$, with $\alpha(x) = nx$.

Theorem 2.3.7 (Mayer-Vietoris). If $X = A \cup B$, with A, B open subsets of X. Then there exists *Mayer-Vietoris boundary homomorphism*

$$\partial_{MV} : \mathrm{H}_{i+1}(X) \to \mathrm{H}_i(A \cap B)$$

for all i, so that the sequence

$$\cdots \longrightarrow \mathrm{H}_{i+1}(X) \xrightarrow{\partial_{MV}} \mathrm{H}_i(A \cap B) \xrightarrow{i_{A*} \oplus i_{B*}} \mathrm{H}_i(A) \oplus \mathrm{H}_i(B) \xrightarrow{j_{A*} - j_{B*}} \mathrm{H}_i(X) \qquad \cdots$$

is exact, where

$$\begin{array}{ccc} A \cap B & \xrightarrow{i_A} & A \\ {\scriptstyle i_B}\downarrow & & \downarrow{\scriptstyle j_A} \\ B & \xrightarrow[j_B]{} & X \end{array}$$

Similarly, we have a homomorphism

$$\partial^*_{MV} : H^i(A \cap B) \to H^{i+1}(X)$$

which makes the sequence

$$\cdots \longrightarrow \mathrm{H}^{i-1}(A \cap B) \xrightarrow{\partial^*_{MV}} \mathrm{H}^i(X) \xrightarrow{j_A^* \oplus -j_B^*} \mathrm{H}^i(A) \oplus \mathrm{H}^i(B) \xrightarrow{i_A^* + i_B^*} \mathrm{H}^i(A \cap B) \xrightarrow{\partial^*_{MV}} \mathrm{H}^{i-1}(X) \longrightarrow \cdots$$

exact.

Remark 2.3.8. (i) The Mayer-Vietoris maps are not induced by maps of spaces, but they are constructed algebraically. Suppose $\sigma \in C_{i+1}(X)$ is a cycle, and suppose we can write $\sigma = \sigma_A + \sigma_B$, where $\sigma_A \in C_{i+1}(A)$ and $\sigma_B \in C_{i+1}(B)$ are chains, and in general, not cycles. Since $\partial\sigma = 0, \partial\sigma_A + \partial\sigma_B = 0$. We define

$$\partial_{MV}(\sigma) = [\partial\sigma_A] \in \mathrm{H}_i(A \cap B)$$

Since $\partial\sigma_A \in C_i(A)$, we must have that $\partial\sigma_A = -\partial\sigma_B \in C_i(A \cap B)$. As $\partial^2 = 0$, $\partial\sigma_A$ is closed and so it represents a class in homology.

This is (correct) intuition, but not a proof.

(ii) The Mayer-Vietoris sequence is natural for maps of pairs. That is, if $X = A \cup B$, $Y = C \cup D$, and $f : X \to Y$ has $f(A) \subseteq C$, $f(B) \subseteq D$, then we have a map of long exact sequences

$$\begin{array}{ccccccccccccc}
\cdots & \longrightarrow & \mathrm{H}_{i+1}(X) & \longrightarrow & \mathrm{H}_i(A \cap B) & \longrightarrow & \mathrm{H}_i(A) \oplus \mathrm{H}_i(B) & \longrightarrow & \mathrm{H}_i(X) & \longrightarrow & \mathrm{H}_{i-1}(A \cap B) & \longrightarrow & \cdots \\
 & & \downarrow{\scriptstyle f_*} & & \downarrow{\scriptstyle f_*} & & \downarrow{\scriptstyle f_* \oplus f_*} & & \downarrow{\scriptstyle f_*} & & \downarrow{\scriptstyle f_*} & & \\
\cdots & \longrightarrow & \mathrm{H}_{i+1}(Y) & \longrightarrow & \mathrm{H}_i(C \cap D) & \longrightarrow & \mathrm{H}_i(C) \oplus \mathrm{H}_i(D) & \longrightarrow & \mathrm{H}_i(Y) & \longrightarrow & \mathrm{H}_{i-1}(C \cap D) & \longrightarrow & \cdots
\end{array}$$

where all the squares commute.

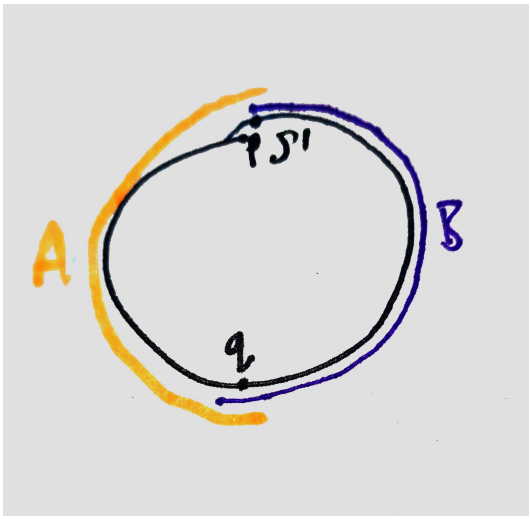
2.4 Examples of Homology calculations

We will prove these later when we use them.

Example 2.4.1

$$\mathrm{H}_*(S^1) = \begin{cases} \mathbb{Z} & * = 0, 1 \\ 0 & \text{otherwise} \end{cases}$$

Proof. Here, $S^1 = A \cup B$, where $A, B \simeq \{*\}$ and $A \cap B \simeq \{p, q\}$.



Using this, we know $H_*(A), H_*(B), H_*(A \cap B)$. For $i \geq 2$, we have

$$\cdots \longrightarrow \underbrace{H_i(A) \oplus H_i(B)}_{=0} \longrightarrow H_i(X) \longrightarrow \underbrace{H_{i-1}(A \cap B)}_{=0} \longrightarrow \cdots$$

and so $H_i(X) = 0$ for all $i \geq 2$. For lowe degrees, we have

$$\begin{array}{ccccccc} \underbrace{H_1(A \cap B)}_{=0} & \longrightarrow & \underbrace{H_1(A) \oplus H_1(B)}_{=0} & \longrightarrow & H_1(X) & & \\ & & & \overset{\partial_{MV}}{\searrow} & & & \\ \underbrace{H_0(A \cap B)}_{\langle p \rangle \oplus \langle q \rangle} & \xrightarrow{\alpha} & \underbrace{H_0(A) \oplus H_0(B)}_{=\mathbb{Z} \oplus \mathbb{Z}} & \longrightarrow & \underbrace{H_0(X)}_{=\mathbb{Z}} & \longrightarrow & 0 \end{array}$$

where $\alpha(n, m) = (n + m, n + m)$. Moreover, we use that S^1 is path connected, and H_0 is generated by any point. Therefore, $H_1(X)$ sits in an exact sequence

$$0 \longrightarrow H_1(X) \xrightarrow{\partial_{MV}} \mathbb{Z}^2 \xrightarrow{\alpha} \mathbb{Z}^2 \xrightarrow{\beta} \mathbb{Z} \longrightarrow 0$$

and so,

$$H_1(X) \simeq \ker(\alpha) = \langle (1, -1) \rangle = \mathbb{Z}(p - q)$$

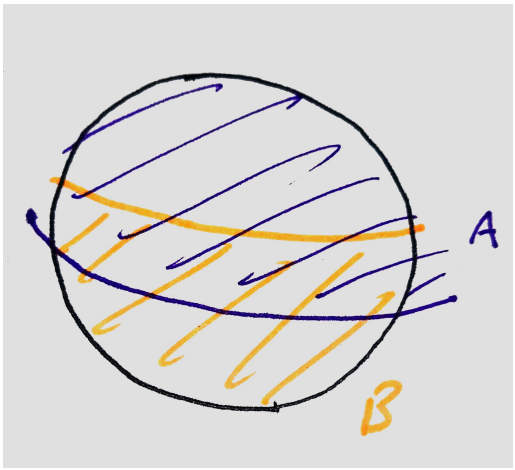
□

Example 2.4.2

$$H^*(S^n) = H_*(S^n) = \begin{cases} \mathbb{Z} & * = 0, n \\ 0 & \text{otherwise} \end{cases}$$

Lecture 5

Proof. We will compute the cohomology. Define $A, B \subseteq S^n$ by



Then A, B are contractible, and $A \cap B$ is homotopic to S^{n-1}. We work inductively in n, using the Mayer-Vietoris sequence. The sequence gives

$$\begin{array}{ccccc} \cdots & \longrightarrow & \mathrm{H}^{i-1}(S^n) & & \\ & \overset{\partial^*_{MV}}{\swarrow} & & & \\ \mathrm{H}^i(S^n) & \longrightarrow & \mathrm{H}^i(*) \oplus \mathrm{H}^i(*) & \longrightarrow & \mathrm{H}^{i-1}(S^{n-1}) \\ & \overset{\partial^*_{MV}}{\swarrow} & & & \\ \mathrm{H}^{i+1}(S^n) & \longrightarrow & \mathrm{H}^{i+1}(*) \oplus \mathrm{H}^i(*) & \longrightarrow & \cdots \end{array}$$

If $i > 0$, we have

$$0 \longrightarrow \mathrm{H}^i(S^{n-1}) \xrightarrow{\partial^*_{MV}} \mathrm{H}^{i+1}(S^n) \longrightarrow 0$$

and so

$$\mathrm{H}^i(S^{n-1}) = \mathrm{H}^{i+1}(S^n)$$

This gives us almost everything inductively. We can assume $n \geq 2$. At the bottom of the sequence, we have

$$\underbrace{\mathrm{H}^0(S^n)}_{=\mathbb{Z}} \longrightarrow \underbrace{\mathrm{H}^0(*) \oplus \mathrm{H}^0(*)}_{=\mathbb{Z}\oplus\mathbb{Z}} \xrightarrow{\alpha} \underbrace{\mathrm{H}^0(S^{n-1})}_{=\mathbb{Z}} \longrightarrow \mathrm{H}^1(S^n) \longrightarrow 0$$

We have shown (on examples sheet 1) that $\mathrm{H}^0(X) = 1$, generated by the 0-dimensional cochain, sending $p \mapsto 1$. As

$$\alpha(p, q) = p + q$$

α is surjective. Thus, we have

$$0 \longrightarrow \mathrm{H}^1(S^n) \longrightarrow 0$$

and so $\mathrm{H}^1(S^n) = 0$. □

Corollary 2.4.3. $\mathbb{R}^n$ is homeomorphic to $\mathbb{R}^m$ if and only if $m = n$.

Proof. Suppose $f : \mathbb{R}^n \to \mathbb{R}^m$ is a homeomorphism. This gives us a homeomorphism

$$f : \mathbb{R}^n \setminus 0 \to \mathbb{R}^m \setminus \{f(0)\}$$

But $\mathbb{R}^n \setminus \{*\}$ is homotopy equivalent to $\S^{n-1}$. Thus f induces an isomorphism in homology

$$\mathrm{H}_i(S^{n-1}) = \mathrm{H}_i(S^{m-1})$$

But by our homology computations, this implies that $n = m$. □

Note in the below we assume $n > 0$.

Definition 2.4.4 (degree)

Suppose $f : S^n \to S^n$ is a continuous map, then f induces a map

$$f_* : \mathrm{H}_n(S^n) \to \mathrm{H}_n(S^n)$$

which is multiplication by some $d \in \mathbb{Z}$. We call $\deg(f) = d$ the *degree* of f.

Remark 2.4.5. Since $\mathrm{H}_n(S^n) = \mathbb{Z}$ which is free, therefore f_* is determined by $f_*(1)$. Moreover, we will need to use the *same* identification of $\mathrm{H}_n(S^n) = \mathbb{Z}$, and so $\deg(f)$ is well defined. Note that if we use different isomorphisms, then $\deg(f)$ is only defined up to a sign.

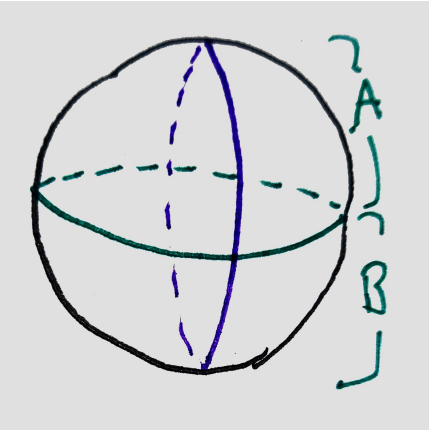
Note that

- if $f \simeq g$, then $\deg(f) = \deg(g)$,
- $\deg(\mathrm{id}) = 1$.
- $\deg(\mathrm{const}) = 0$, since the constant map factors through $\{*\}$, and so the induced map on homology factors through $\mathrm{H}^n(*) = 0$.

Lemma 2.4.6. If $A \in O(n+1)$, then A acts on S^n, and we have

$$\deg(A) = \det(A) \in \{\pm 1\}$$

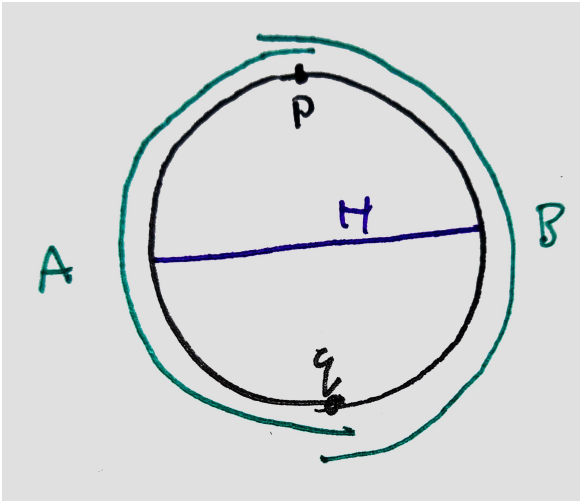
Proof. The group $O(n+1)$ has two path components. By homotopy invariance of degree, any A is homotopic to id_{S^n}, or a reflection in an equatorial hyperplane H.



The reflection preserves $A, B, A \cap B$, and H intersects $A \cap B = S^{n-1}$ at two points. The Mayer-Vietoris sequence gives

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathrm{H}_n(S^n) & \xrightarrow{\cong} & \mathrm{H}_{n-1}(S^{n-1}) & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle A_*} & & \downarrow{\scriptstyle A_*} & & \\ 0 & \longrightarrow & \mathrm{H}_n(S^n) & \xrightarrow[\cong]{} & \mathrm{H}_{n-1}(S^{n-1}) & \longrightarrow & 0 \end{array}$$

The two vertical maps are the same. Therefore it suffices to consider the result on S^1.



Recall our computation of $H_1(S^1)$ has

$$0 \longrightarrow \mathrm{H}_1(S^1) \longrightarrow \mathrm{H}_0(p \sqcup q) \xrightarrow{\alpha} \mathrm{H}_0(A) \oplus \mathrm{H}_0(B) \longrightarrow \cdots$$

where $\alpha(u, v) = (u + v, u + v)$. Hence $H_1(S^1)$ is generated by $(1, -1)$, which we can think of as $p - q$. But in this case, reflection in H swaps p and q, and so it acts on $H^1(S^1)$ by multiplication by -1. □

Corollary 2.4.7. The antipodal map $a_n : S^n \to S^n$ has degree $(-1)^{n+1}$.

Proof. a_n is orthogonal. Or equivalently, it is a composition of $n + 1$ reflections, and we have that

$$\deg(fg) = \deg(f)\deg(g)$$

□

Corollary 2.4.8. If $f : S^n \to S^n$ has no fixed point, then f is homotopic to the antipodal map.

Proof. In fact, we will see that $f(x) \neq g(x)$ for all $x \in S^n$, then $f \simeq a_n \circ g$. Taking $g = \mathrm{id}$ gives the result. Consider the map

$$x \mapsto \frac{tf(x) - (1-t)g(x)}{\left\| tf(x) - (1-t)g(x) \right\|}$$

for $0 \leq t \leq 1$. Note that the denominator never vanishes. This then defines a homotopy from f to $-g$. □

Using this, the degree of any map without fixed points is $(-1)^{n+1}$. Another corollary is that if $f(x) \neq -x$ for all $x \in S^n$, then $f \simeq a \circ a = \mathrm{id}$.

Definition 2.4.9 (vector field on S^n)
A *vector field on* S^n is a map $v : S^n \to \mathbb{R}^{n+1}$, such that for all $x \in S^n$,

$$\langle x, v(x) \rangle = 0$$

Proposition 2.4.10 (Hairy ball theorem). S^n has a nowhere vanishing vector field if and only if n is odd.

Proof. If $n = 2k - 1$,

$$v(x_1, y_1, \ldots, x_k, y_k) = (-y_1, x_2, \ldots, -y_k, x_k)$$

works.

Now suppose a nowhere vanishing vector field $v : S^n \to \mathbb{R}^{n+1}$ exists. By normalising, i.e. considering

$$\frac{v(x)}{\|v(x)\|}$$

we can consider $v : S^n \to S^n \subseteq \mathbb{R}^{n+1}$. Now consider the family

$$v_t(x) = \cos(t)x + \sin(t)v(x)$$

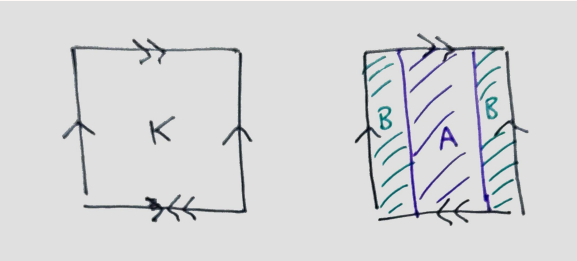
for $0 \leq t \leq \pi$, which has unit length as we assumed $v(x)$ and x are orthogonal. This family has $v_0 = \mathrm{id}$, and $v_\pi = a_n$. Thus, we have that

$$1 = \deg(\mathrm{id}) = \deg(a_n) = (-1)^{n+1}$$

□

Example 2.4.11 (Klein bottle)

Recall the Klein bottle is the result if we glue two Möbius strips together along their common boundary S^1. That is, we define K by the gluing pattern



In this case, we can write $K = A \cup B$, where A, B are Möbius bands, i.e. $A \simeq S^1 \simeq B$. Moreover, $A \cap B \simeq S^1$ as well. The interesting part of the Mayer-Vietoris sequence for homology is as follows:

$$0 \longrightarrow H_2(K) \longrightarrow H_1(A \cap B) \xrightarrow{\psi} H_1(A) \oplus H_1(B) \longrightarrow H_1(K) \longrightarrow H_0(A \cap B) \xrightarrow{\alpha} H_0(A) \oplus H_0(B)$$

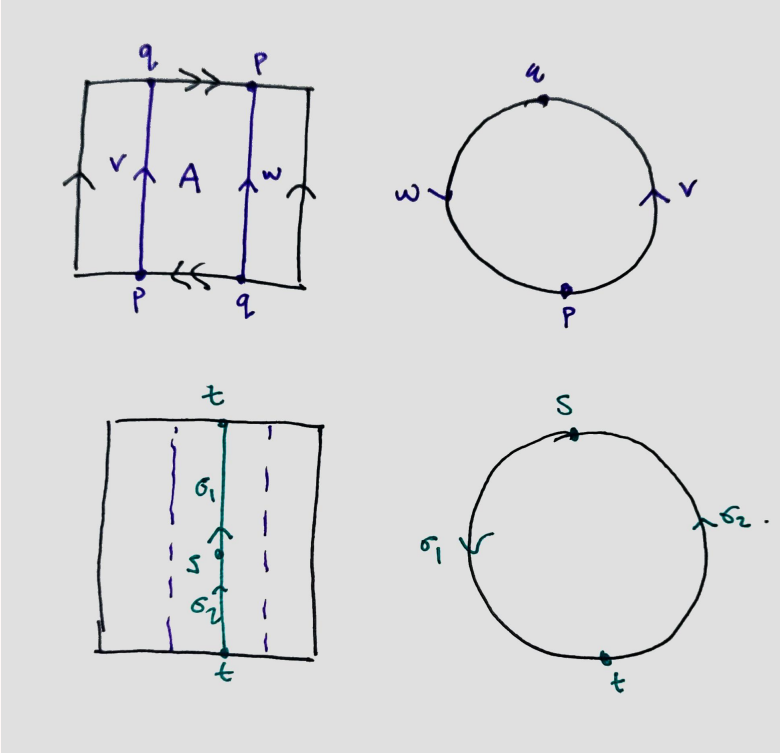
Recall $\alpha(1) = (1, 1)$, and so α is injective, and the sequence becomes

$$0 \longrightarrow H_2(K) \longrightarrow H_1(A \cap B) \xrightarrow{\psi} H_1(A) \oplus H_1(B) \longrightarrow H_1(K) \longrightarrow 0$$

By exactness, $H_1(K) \cong (\mathbb{Z} \oplus \mathbb{Z})/\operatorname{im}(\psi) = \operatorname{coker}(\psi)$ and $H_2(K) \cong \ker(\psi)$.

Claim 2.4.12. $\psi(1) = (2, 2)$, and so $H_2(K) = 0$, and $H_1(K) = \mathbb{Z} \oplus \mathbb{Z}/2$.

Proof of claim. We will use the decomposition of K into two Möbius bands.



In this case, $H_1(A \cap B) = \mathbb{Z}\langle v + w \rangle$, and $H_1(A) = \mathbb{Z}\langle \sigma_1 + \sigma_2 \rangle$. At chain level, there exists a collection of 2-simplices in A, with boundary $c - (\sigma_1 + \sigma_2)$. Therefore, $v \mapsto \sigma_1 + \sigma_2$ and $w \mapsto \sigma_1 + \sigma_2$. With this, we have that (by symmetry),

$$\psi(1) = (2, 2)$$

□

Remark 2.4.13. (i) In cohomology, $H^2(K) = \mathbb{Z}/2$ and $H^1(K) = \mathbb{Z}$. That is, the homology and the cohomology are not the same.

(ii) We defined $C_i(X) = \left\{\sum_{\text{finite}} a_j \sigma_j \mid a_j \in \mathbb{Z}, \sigma_j \text{ is an } i\text{-simplex}\right\}$. But we can also define

$$C_i(X; G) = \left\{\sum_{\text{finite}} a_j \sigma_j \;\middle|\; a_j \in G, \sigma_j \text{ is an } i\text{-simplex}\right\}$$

for any abelian group G. ∂ is defined as before, and we get a chain complex $C_*(X; G)$ and homology $H_*(X; G)$. If we instead computed

$$H_*(K; \mathbb{Z}/2) = \begin{cases} \mathbb{Z}/2 & * = 0 \\ \mathbb{Z}/2 \oplus \mathbb{Z}/2 & * = 1 \\ \mathbb{Z}/2 & * = 2 \end{cases}$$

since $\psi = 0$ if we used $\mathbb{Z}/2$ coefficients.

Interlude on relative homology

Some of the questions on sheet 1 concern *relative homology*. If $A \subseteq X$ is a subspace, then any simplex $\sigma : \Delta^1 \to A$ is a simplex in X. Moreover, if σ is a simplex in A, so are all of its boundary faces. Thus, the natural map induced by the inclusion map,

$$C_i(A) \hookrightarrow C_i(X)$$

defines a chain map. Using this, $C_*(A)$ is a *subcomplex* of $C_*(X)$. In particular, we can define the *quotient complex*,

$$C_i(X, A) = \frac{C_i(X)}{C_i(A)}$$

and this is a chain complex, with ∂ induced by the boundary operator on $C_i(X)$. Another way of thinking about $C_i(X, A)$ is that it is the free abelian group on i-simplices in X not wholly contained in A. The homology of $C_*(X, A)$, denoted $\mathrm{H}_*(X, A)$ is called *relative homology*.

We have a long exact sequence

$$\cdots \longrightarrow \mathrm{H}_i(A) \longrightarrow \mathrm{H}_i(X) \longrightarrow \mathrm{H}_i(X, A) \xrightarrow{\partial} \mathrm{H}_{i-1}(A) \longrightarrow \cdots$$

If A and X are 'well behaved', then $\mathrm{H}_i(X, A) \cong \mathrm{H}_i(X/A)$ for $i > 0$, where X/A is the quotient space where we quotient A into a point.

2.5 Homotopy invariance

Recall if $f, g : X \to Y$ are homotopic, we would like to show that the induced maps on homology and cohomology are the same. Recall also that the maps are induced by (co)chain maps on (co)chain complexes.

Definition 2.5.1 (chain homotopic)

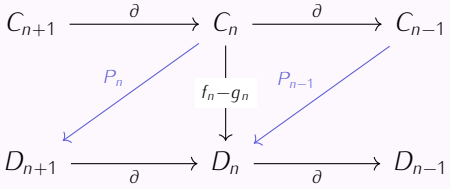
Let C_*, D_* be chain complexes. Let $f_*, g_* : C_* \to D_*$ be chain maps. We say that they are *chain homotopic* if there exists maps

$$P_n : C_n \to D_{n+1}$$

for all n, such that

$$P_{n-1} \circ \partial + \partial \circ P_n = f_n - g_n$$

for all n.



Lemma 2.5.2. Suppose $f_*, g_* : C_* \to D_*$ are chain homotopic. Then they induce the same map on homology.

Proof. Let $\alpha \in C_i$ be a cycle, that is, $\partial\alpha = 0$. By definition, $f_*[\alpha] = [f(\alpha)]$. Consider

$$f_i(\alpha) - g_i(\alpha) = (f_i - g_i)(\alpha) = (\partial P + P\partial)(\alpha) = \partial(P\alpha)$$

But this is in the image of ∂, and so the corresponding homology classes are the same. □

Lecture 7

Exercise: Chain homotopy is an equivalence relation on chain maps $C_* \to D_*$.

Theorem 2.5.3 (homotopy invariance of homology)**.** If $f, g : X \to Y$ are homotopic, then the induced maps on homology are the same.

Proof. Let $i_j : X \hookrightarrow X \times \{j\}$ be the natural map, $F : X \times I \to Y$ the homotopy between f, g. But then

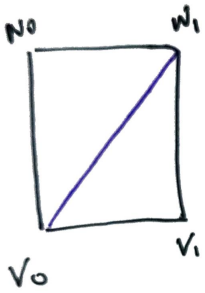
$$f_* = (F \circ i_0)_* = F_* \circ i_{0*}$$

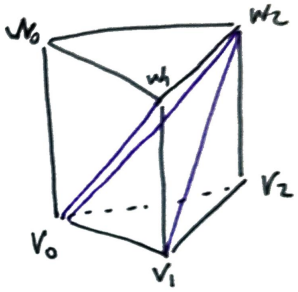
Therefore, it suffices to show that i_{0*} and i_{1*} are chain homotopic.

Thus, we want

$$P : C_n(X) \to C_{n+1}(X \times [0, 1])$$

such that $\partial P + P\partial = i_{0*} - i_{1*}$. P is a *prism operator* from a universal way of decomposing $\Delta^n \times [0, 1]$ into a finite collection of $n + 1$-simplices.





Consider the ordered collections$[v_0, \dots, v_i, w_i, \dots, w_n]$, where $[v_0, \dots, v_n] = \Delta^n \times \{0\}$ and $[w_0, \dots, w_n] = \Delta^n \times \{1\}$. We define P by

$$P(\sigma) = \sum_{i=0}^{n} (-1)^i (\sigma \times I)|_{[v_0,\dots,v_i,w_i,\dots,w_n]}$$

Claim 2.5.4. $\partial P + P\partial = i_{i*} - i_{0*}$. That is,

$$\partial P = i_{1*} - i_{0*} - P\partial$$

i.e. "boundary of prism is the top and bottom faces, as well as the prism of the boundary".

Proof.

$$\partial P(\sigma) = \sum_{j \le i} (-1)^i (-1)^j (\sigma \times I)|_{[v_0,\dots,\widehat{v_j},\dots,v_i,w_i,\dots,w_n]} + \sum_{j \ge i} (-1)^i (-1)^{j+1} (\sigma \times I)|_{[v_0,\dots,v_i,w_i,\dots,\widehat{w_j},\dots,w_n]}$$

The terms when $j = i$ cancel, except for

$$(\sigma \times I)|_{[w_0,\dots,w_n]} - (\sigma \times I)|_{[v_0,\dots,v_n]}$$

which are the top and bottom faces. The terms with $i > j$ and $i < j$ comprise, up to a sign, $P(\partial\sigma)$. Just compute (in a dark room with some gin). □

□

Remark 2.5.5. If C^*, D^* are cochain complexes, cochain maps f^*, g^* are cochain homotopic if we have $P^i : C^i \to D^{i-1}$, such that

$$\partial P + P\partial = f^* - g^*$$

It is easy to check that in this case, f^* and g^* induce the saem map $\mathrm{H}^*(C^*) \to \mathrm{H}^*(D^*)$.

With this, our prism operator $P : C_n(X) \to C_{n+1}(X \times [0,1])$ has a dual

$$P^* : C^{n+1}(X \times [0,1]) \to C^n(X)$$
$$P^*(f) = f \circ P$$

Dualising everything, the relation $\partial P + P\partial = i_{1*} - i_{0*}$ becomes

$$\partial P^* + P^*\partial = i_1^* - i_0^*$$

and thus homotopic maps induce the same map on cohomology.

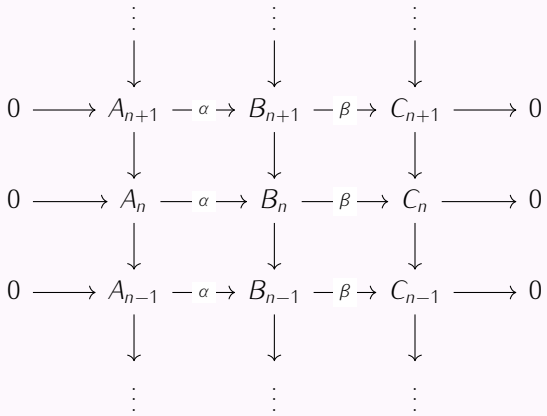
2.6 Snake lemma

Definition 2.6.1

A *short exact sequence of chain complexes*

$$0 \longrightarrow A_* \longrightarrow B_* \longrightarrow C_* \longrightarrow 0$$

is a diagram



such that the squares commute (i.e. α, β are chain maps) and the rows are exact.

Proposition 2.6.2 (Snake lemma). Suppose we have a SES $0 \to A_* \to B_* \to C_* \to 0$ of chain complexes, then we have a long exact sequence of homology

$$\cdots \longrightarrow \mathrm{H}_{n+1}(C_*) \xrightarrow{\delta} \mathrm{H}_n(A_*) \xrightarrow{\alpha_*} \mathrm{H}_n(B_*) \xrightarrow{\beta_*} \mathrm{H}_n(C_*) \xrightarrow{\delta} \mathrm{H}_{n-1}(A_*) \longrightarrow \cdots$$

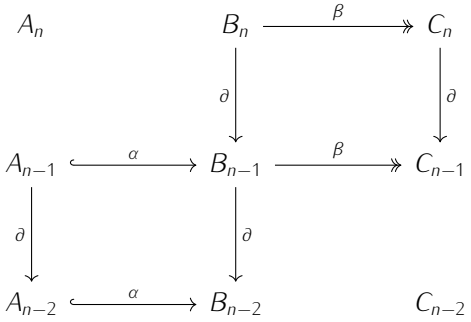
Proof. We will construct δ, but leave all other details. Suppose $c_n \in C_n$ is a cycle, representing a homology class $[c_n] \in H_n(C^*)$. Horizontal exactness means $c_n = \beta(b_n)$ for some $b_n \in B_n$. But

$$\beta(\partial b_n) = \partial\beta(b_n) = \partial c_n = 0$$

Hence $\partial b_n \in \ker(\beta) = \mathrm{im}(\alpha)$, by horizontal exactness. Thus, we have $a_{n-1} \in A_{n-1}$ such that $\alpha(a_{n-1}) = \partial b_n$. In this case,

$$\alpha(\partial a_{n-1}) = \partial\alpha(a_{n-1}) = \partial^2 b_n = 0$$

But α is injective, and so $\partial a_{n-1} = 0$, and this represents a cycle.



We define $\delta([c_n]) = [a_{n-1}]$.

To complete the proof:

1. Check that δ is independent of the choices of c_n, b_n, a_{n-1}.
2. δ is a homomorphism.
3. Check that the sequence is exact.

□

Example 2.6.3

Recall if G is an abelian group, we introducted $C_*(X; G)$. If we started with an SES of abelian groups

$$0 \longrightarrow G_1 \longrightarrow G_2 \longrightarrow G_3 \longrightarrow 0$$

then there is an associated SES of chain complexes

$$0 \longrightarrow C_*(X; G_1) \longrightarrow C_*(X; G_2) \longrightarrow C_*(X; G_3) \longrightarrow 0$$

The homomorphism

$$\mathrm{H}_n(X; G_3) \to \mathrm{H}_{n-1}(X; G_1)$$

are often called *Bockstein homomorphisms*.

Example 2.6.4

If $A \subseteq X$ is a subspace, then we defined relative homology, using the SES

$$0 \longrightarrow C_*(A) \longrightarrow C_*(X) \longrightarrow C_*(X, A) \longrightarrow 0$$

The associated long exact sequence includes a boundary map

$$\delta : \mathrm{H}_n(X, A) \to \mathrm{H}_{n-1}(A)$$

and we have a LES of the pair (X, A).

2.7 Excision

Theorem 2.7.1 (Excision). Let X be a topological space, $A \subseteq X$ a subspace, $Z \subseteq X$ a subspace with $\mathrm{Cl}(Z) \subseteq \mathrm{Int}(A)$. Then the inclusion

$$i : (X \setminus \mathbb{Z}, A \setminus Z) \to (X, A)$$

induces an isomorphism on relative homology

$$i_* : \mathrm{H}_n(X \setminus Z, A \setminus Z) \to \mathrm{H}_n(X, A)$$

Lemma 2.7.2 (five lemma). Suppose we have a diagram

$$\begin{array}{ccccccccc} A & \longrightarrow & B & \longrightarrow & C & \longrightarrow & D & \longrightarrow & E \\ \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle \beta} & & \downarrow{\scriptstyle \gamma} & & \downarrow{\scriptstyle \delta} & & \downarrow{\scriptstyle \varepsilon} \\ A' & \longrightarrow & B' & \longrightarrow & C' & \longrightarrow & D' & \longrightarrow & E' \end{array}$$

where the rows are exact, $\alpha, \beta, \delta, \varepsilon$ are isomorphisms, then so is δ.

Corollary 2.7.3. If $f : (X, A) \to (Y, B)$ is a map of pairs, and the maps $f_* : \mathrm{H}_*(X) \to |HH_*(Y)$ and $f_* : \mathrm{H}_*(A) \to \mathrm{H}_*(B)$ are isomorphisms, then so are the induced maps on relative homology

$$f_* : \mathrm{H}_*(X, A) \to \mathrm{H}_*(Y, B)$$

Proof. Follows from the five lemma and the LES of relative homology. □

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